

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	$I_1 = I_2 = I_3$ or two of $I_1 = I_2$, $I_2 = I_3$, $I_1 = I_3$	1			1		
		(ii)	$V_1 + V_2 = V_3$	1			1		
	(b)	(i)	Correct application of any potential divider equation, e.g. $\frac{V_{out}}{12\text{ V}} = \frac{32\ \Omega}{48\ \Omega}$ or current = $\frac{12\text{ V}}{48\ \Omega}$ [= 0.25 A] (1) $V_{out} = 8.0\text{ [V]}$ if it follows clearly from some correct working (1)		2		2	2	
		(ii)	$V_{out} = \left[12\text{ V} \times \frac{12\ \Omega}{48\ \Omega}\right] = 3.0\text{ [V]}$ or $[0.25\text{ ecf} \times 12] = 3.0\text{ [V]}$		1		1	1	
	(c)		R_A and R_B each have 2.0 V across them and the 20 Ω resistor has 8.0 V across it (1) $R_A = 5\ [\Omega]$, $R_B = 5\ [\Omega]$ (1) Alternative 1 ($V = IR$) award 2 marks for: $I = \frac{2}{R_B} = \frac{10}{20+R_B}$ leads to $R_B = 5\ [\Omega]$ Then $I = \frac{10}{25} = 0.4\text{ [A]}$ Then $R_A = \frac{2}{0.4} = 5\ [\Omega]$ Alternative 2 ($\frac{V_{out}}{V_{in}} = \frac{R}{R_T}$) For $R_B + 20$: $\frac{10}{12} = \frac{R_B+20}{R_A+20+R_B}$ and for R_B : $\frac{2}{12} = \frac{R_B}{R_A+20+R_B}$ (1) Multiplying by 5 gives $\frac{10}{12} = \frac{5R_B}{R_A+20+R_B}$ and so $5R_B = R_B + 20$ $R_B = 5\ [\Omega]$ and with a sub back in $R_A = 5\ [\Omega]$ (1)			2	2		
			Question 2 total	2	3	2	7	3	0